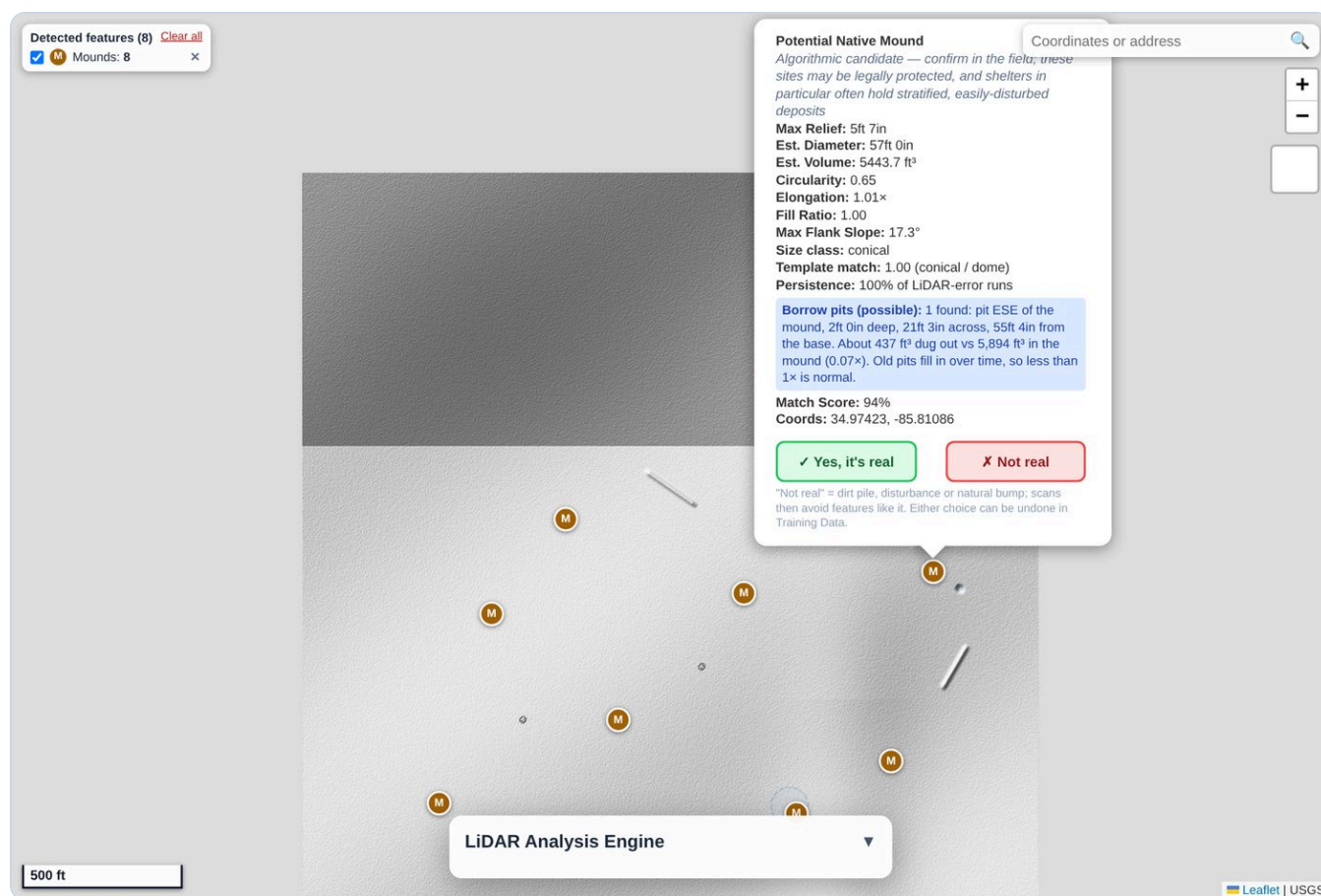


LiDAR Analysis Engine User Guide & Feature List

Finding mounds, caves, rockshelters and camp sites in bare-earth LiDAR, and teaching the app to find more.



Covers the version of *index.html* as of 2 October 2026. Screenshots use sample terrain, so map backgrounds are plain grey; on your device the satellite, street or topo map shows underneath.

Contents

Part 1 — Feature list

- 1.1 Map, search and navigation
- 1.2 Terrain data
- 1.3 Terrain display
- 1.4 Scan modes
- 1.5 Mound detection
- 1.6 Results, legend and exports
- 1.7 Training the app
- 1.8 Managing training data
- 1.9 Deep learning (experimental)
- 1.10 Saving and backups

Part 2 — Walk-through

- 2.1 Opening the app and the screen layout
- 2.2 Going to your area
- 2.3 Loading terrain
- 2.4 Looking at the terrain

- 2.5 Scanning
- 2.6 Reading and answering results
- 2.7 Checking a spot by hand
- 2.8 Outlining mounds and look-alikes
- 2.9 Teaching the scan your mounds
- 2.10 Reviewing and editing training data
- 2.11 Checking training data for issues
- 2.12 Deep learning, start to finish
- 2.13 Exporting for the field
- 2.14 Backing everything up
- 2.15 Using it on a phone
- 2.16 Batch pipeline (desktop)

Part 3 — Reference

- 3.1 Settings reference
- 3.2 Troubleshooting
- 3.3 Responsible use

The short version. Search for your area → **Download Entire Map View** → pick a scan mode → **Scan Downloaded DEM** → tap each result and answer **✓ Yes, it's real** or **✗ Not real**. Every answer teaches the app, and everything is saved in your browser. Back it up with **Export Training Data** and **Save deep-learning model**.

PART 1

Feature list

Everything the app can do, grouped by area. Part 2 shows how to use each of them.

1.1 Map, search and navigation

Basemaps	Street Map, Satellite Imagery and USGS Topo, switched from the layers button at the top right.
Overlay layers	LiDAR Hillshade, Slope, Settlement Likelihood, Detected Features, and the FEMA 100-year Flood Hazard map (for checking floodplains by eye).
Search	A box at the top right of the map (a  button on a phone) and one in the panel. Reads decimal degrees in either order, degrees-minutes-seconds, degrees with decimal minutes, UTM (e.g. 16S 608297 3870858) and pasted map links. Anything else is looked up as an address or place name (OpenStreetMap, then Esri), with a list when there are several matches. A red pin shows the spot in decimal degrees, DMS and UTM.
Scale bar	Bottom left; feet, switching to miles when zoomed out.
Collapsible panel	The settings panel folds down to its title bar ( / ); it folds itself away on a phone when you search, draw or review results on the map.

1.2 Terrain data

Download Entire Map View	Downloads the USGS 3DEP bare-earth DTM (ground-classified LiDAR, canopy removed) for everything in view, in tiles fetched in parallel and stitched into one seamless grid. Auto resolution gives full 1 m LiDAR whenever the view fits the memory budget; 0.5 m to 10 m can be chosen. Memory budget from 16 Mpx (phones) to 256 Mpx (desktop).
Load Local DTM	Opens your own GeoTIFFs, including the original USGS 1 m DEM tiles in UTM (NAD83 / NAD83(2011) / WGS84), converted in the browser. Several tiles can be opened at once and are merged; only the part under the map view is read, so full 10 × 10 km tiles work on a phone.
What data is here?	Tap the map: shows which source raster is under that point and its cell size, so you know whether small features can be trusted.
1 m minimum	<i>Require ≤ 1 m/px DTM before scanning blocks scans on coarse data (untick to allow).</i>
Load Pipeline Results	Opens candidate GeoJSON from the desktop batch pipeline for review, including in 3D.

1.3 Terrain display

Hillshade	Drawn on the graphics card with smooth (bicubic) interpolation and per-pixel lighting. Azimuth, Altitude, Z-Exaggeration and opacity sliders.
Multi-Directional Lighting	Four lights around the compass, so a mound's shaded side doesn't disappear.
Slope layer	True slope angle (green flat → dark red 60°+), good for bluff lines and cliff faces.
Terrain Smoothing	Off (raw data, the default), Auto or Strong. Display only: removes flat triangle facets left by sparse ground returns under forest and blocky coarse-source data, while keeping real edges sharp. Scans always use the raw DTM.
3D Fly-Through	Renders the DTM in 3D with every detected feature in its mode colour and pipeline context in amber.

1.4 Scan modes

Mode	What it finds
Phase 2: Long-Term Habitation (Base Camps)	Flat dirt benches at the foot of south/southeast-facing bluffs (TPI → aspect → Local Relief Model), near year-round water and above the 100-year floodplain; flags possible middens under the drip line.
Phase 3: Transient Hunt Camps	Minor rock outcrops next to natural choke points (saddles, ridge gaps, narrow canyon funnels), with a viewshed of the valley floor and stream crossings they overlook (<i>Show viewshed on map</i>).
Phase 4: Deep Ceremonial Karst	Large, deep closed depressions (possible collapsed karst windows), sinking streams, cross-referenced against a cave survey you load (GeoJSON/CSV) to find sinks that open into caves long enough to have a dark zone.
Horizontal Cave Entrances	The settings walk-in cave mouths open into: gentle ground or a talus apron at a cliff foot, usually with a cove head or a spring-cut stream head.
Caves / Sinkholes	Closed depressions by Priority-Flood fill, with bowl-shape checks and creek-pool rejection.
Native Mounds	Earthen mounds with the iMound method, persistence testing and template matching (see 1.5), with borrow-pit detection.
Rockshelters / Overhangs	Flat ground at the base of a real cliff (the overhang itself can't be seen in bare-earth data).
Settlement Likelihood	A weighted surface of water proximity, flood safety, flat ground and nearness to known sites, with the top areas highlighted.
Scan All	Runs every mode in turn, each with its own settings; all results stay on the map together.
Auto-Detect Best Settings	Measures the area (LiDAR noise, slopes, bluff heights, drainage, size) and sets the sliders to suit it, after each download or on demand. Every slider stays adjustable.

1.5 Mound detection

- **iMound** (Freeland et al. 2016): the terrain is detrended at several scales and flipped; only closed highs count, so hillside bulges and ridge noses don't. A second *spur pass* separates a mound from the spur it sits on.
- **Hillside measurement**: on tilted ground the height, width and shape are measured above a curved surface fitted to the surrounding slope, so a mound on a mountainside (such as Cottonpatch Mound at Russell Cave) isn't measured from its uphill rim.
- **Persistence**: each candidate is re-tested with random LiDAR-grade error added and must survive most runs.
- **Template matching** against an ideal conical dome and a flat-topped platform; each mound is labelled conical or platform.
- **Mound Shape presets**: All Shapes (recommended), Conical, Platform, Broad/Any, Custom, and **Learned from my confirmed mounds** (after 5 confirmations).
- **Borrow pits**: pits or a ditch around the mound, with dug-out volume compared to the mound, a water flag for ponded pits, and a sinkhole warning. Setting: look for them (rank mounds with pits higher), show only mounds with pits, or don't look.
- **Rugged-ground and slope rules** keep boulders, ledge knobs and anything on ground steeper than 20° out.

1.6 Results, legend and exports

Markers and legend	One coloured, lettered marker per mode; the legend at the top left counts each type and shows/hides it (× clears that type).
Result popups	Measurements, match score, warnings, borrow pits, and two well-separated buttons: ✓ Yes, it's real and ✗ Not real .
Click-to-inspect	Tap anywhere: elevation, the last scan's value there, <i>What data is here?</i> , and <i>This is a Cave/Sinkhole</i> , <i>This is a Mound</i> , <i>This is a Shelter</i> , <i>Confirm Known Site</i> .
Max results per mode	Candidates are ranked by Match Score; nearby hits count as one; only the best are shown (40 by default).
Exports	GPX waypoints (named file, Garmin-style symbols per type), GeoJSON candidates, Shapefile (.zip) and the analysis raster as GeoTIFF.

1.7 Training the app

- **Confirm / Not real** on any result. Confirmed mounds become permanent waypoints that every later scan shows, with the size you recorded. "Not real" removes the result and makes scans skip that spot and skip or rank lower anything that looks like it.
- **Avoid look-alikes** strictness: Light (90% similar), Normal (80%, default), Strict (65%), Off (only rank lower).

- **Outline a Mound or Look-alike:** tap around a mound's base; the app measures it against the surrounding ground and records it as a mound or a look-alike (tree throw, boulder, spoil pile, knob, road/berm).
- **Learned settings:** once 5 confirmed mounds have scanner measurements, the size and shape ranges are learned from them and kept up to date.
- **Known settlement sites** with the evidence you have; cave/shelter **burial or ritual significance** marks.

1.8 Managing training data

Undo last training change	Reverses your latest actions one at a time (up to 30), even after a reload.
Review Training Data	Everything you've taught the app, newest first, with Go to and Remove.
Show Training Data on Map	All items at once: confirmed (green ✓), not real (red ✗), outlines, settlement sites. Tap one to Edit (real/not real, height, width, name), Move (drag) or Remove.
Check training data for issues	Finds records that may be wrong — sizes no mound has, missing sizes, impossible steepness, terrain that doesn't match, a mound confirmed twice, a mound next to a "Not real" mark, and (with a trained network) labels it strongly disagrees with. Each has Go to & edit , It's accurate and Remove. A warning in the panel shows how many there are.

1.9 Deep learning (experimental)

- A small neural network trained **in your browser** on your own confirmed mounds, "Not real" marks, outlines and random ground.
- Sees 40 m squares of terrain in three views (relief above the tilted ground, fine relief, steepness); mounds wider than 35 m are left to the regular scan.
- Terrain samples are collected whenever a DTM is loaded; **Show them on the map** leads you to items still missing one, with **Download this area**.
- Reports a held-out accuracy check after training; scans the area in view and double-checks every candidate 8 ways; cross-checks with the regular mound scan; flags candidates smaller than any mound.
- **Save / Load deep-learning model** as one file, including samples and training data.

1.10 Saving and backups

- Training data is saved automatically in two places in the browser (localStorage and IndexedDB), and the app asks the browser to keep it permanently.
- The panel counts confirmations not yet backed up, with **Back up now**; where supported, a backup file can be kept updated automatically.
- **Export / Import Training Data** (import merges, never loses newer work); **Export / Import outlines** as GeoJSON.

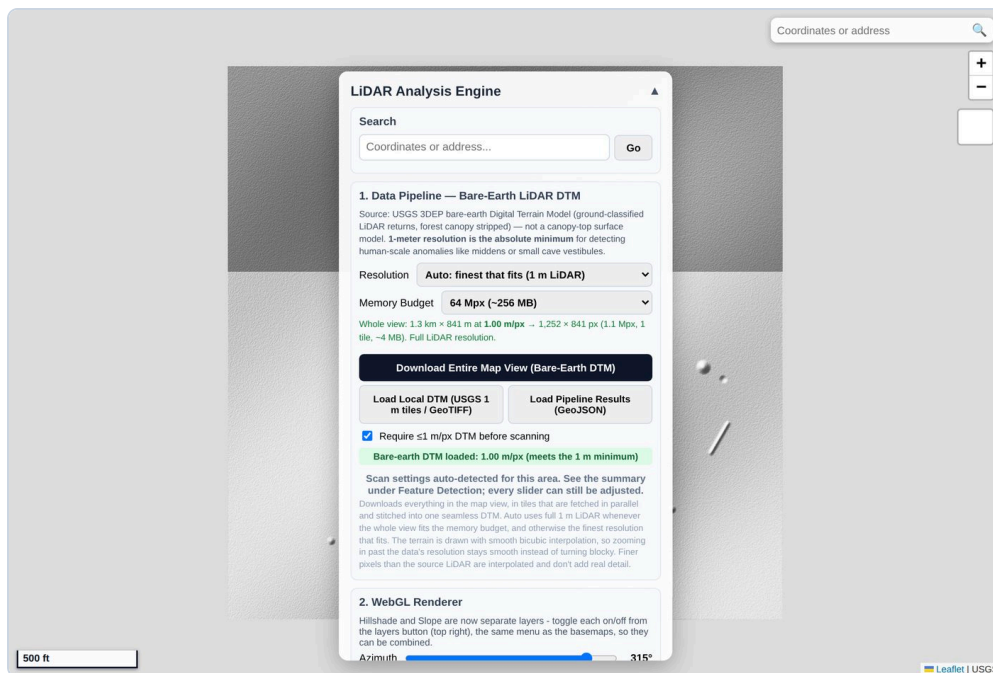
PART 2

Walk-through

This part follows a typical session from start to finish. You don't need every step every time: most sessions are *go to area* → *download* → *scan* → *answer results*.

2.1 Opening the app and the screen layout

Open **index.html** in a browser (Chrome, Edge, Safari or Firefox), or the GitHub Pages address if it's published there. Nothing needs installing. Use the same address every time: your training data is stored for that exact address.

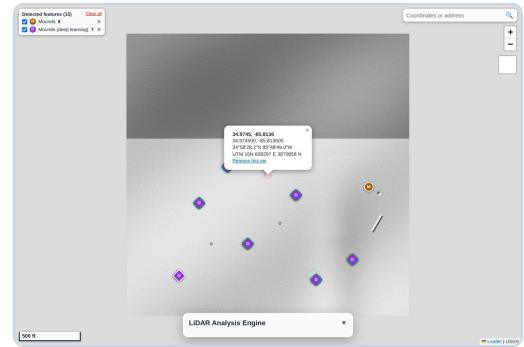


The app after loading terrain. The panel (centre) holds every setting; the map fills the screen.

- **Panel** (centre bottom): sections 1–6 plus Search. Tap the title bar or ▲/▼ to fold it away.
- **Search box** (top right) and **zoom buttons**; the square below them is the **layers** button (basemaps and overlays).
- **Legend** (top left) appears once there are results.
- **Scale bar** (bottom left); **status messages** appear in the panel under section 1 and in each section.

2.2 Going to your area

- 1 Type into the search box at the top right (or the panel's Search box) and press Enter or .
- 2 Use coordinates in almost any form — 34.9745, -85.8136, 34°58'28"N 85°48'49"W, 16S 608297 3870858 — or an address or place name. A pasted Google/Apple map link works too.
- 3 A red pin marks the spot and shows it in decimal degrees, DMS and UTM. *Remove this pin* clears it.
- 4 Zoom so the area you want to study fills the screen: the download covers exactly what's in view.

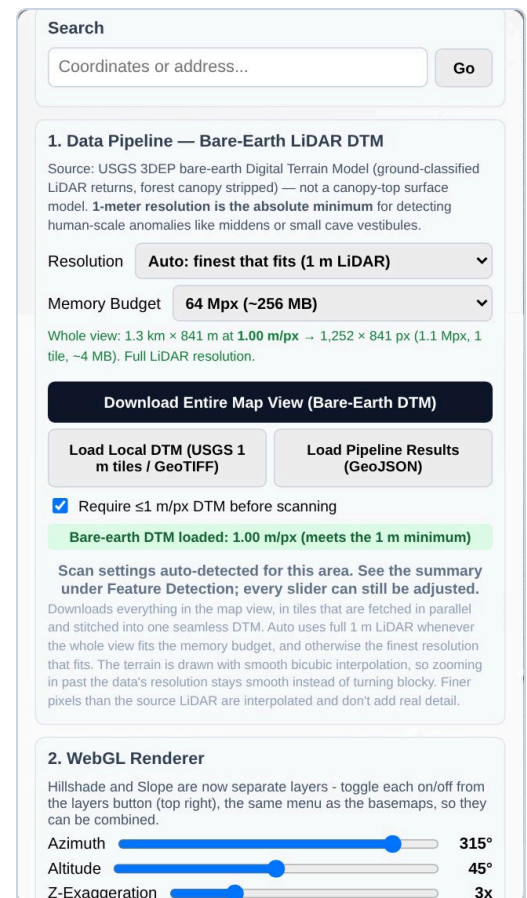


A search result pin with its coordinates in three formats.

2.3 Loading terrain

- 1 Open section **1. Data Pipeline**. Leave **Resolution** on *Auto* and pick a **Memory Budget** (16–32 Mpx on a phone, 64 Mpx or more on a computer). The green line under them shows the size and resolution the view will download at.
- 2 Press **Download Entire Map View (Bare-Earth DTM)**. A progress bar runs; **Cancel download** stops it.
- 3 When it finishes, the status confirms the resolution, e.g. "Bare-earth DTM loaded: 1.00 m/px (meets the 1 m minimum)", and the hillshade appears.

If the app warns that the source data here looks coarser (for example ~4 m), the area isn't covered by 1 m LiDAR in the download service. Get the original USGS 1 m tiles from apps.nationalmap.gov/downloader (Elevation Products (3DEP) → 1 meter DEM) and open them with **Load Local DTM**. Select several tiles at once to merge them.

A screenshot of the '1. Data Pipeline — Bare-Earth LiDAR DTM' settings panel. It includes a search bar at the top. Below it, the 'Resolution' is set to 'Auto: finest that fits (1 m LiDAR)' and the 'Memory Budget' is set to '64 Mpx (~256 MB)'. A green status bar indicates 'Bare-earth DTM loaded: 1.00 m/px (meets the 1 m minimum)'. There are buttons for 'Download Entire Map View (Bare-Earth DTM)', 'Load Local DTM (USGS 1 m tiles / GeoTIFF)', and 'Load Pipeline Results (GeoJSON)'. A checkbox 'Require ≤1 m/px DTM before scanning' is checked. Below this, there's a section for '2. WebGL Renderer' with sliders for 'Azimuth' (315°), 'Altitude' (45°), and 'Z-Exaggeration' (3x).

Section 1: resolution, memory budget, download and local files.

Is the data good enough here? Tap the map and choose **What data is here?** to see which source raster is under that point and its cell size.

2.4 Looking at the terrain

- **Azimuth / Altitude** move the light; **Z-Exaggeration** makes low features stand out. Turn on **Multi-Directional Lighting** so a mound looks like a mound from every side.
- **Terrain Smoothing**: leave *Off* to see the raw data; try *Auto* where forest canopy left triangular facets. It only changes the picture, never the scans.
- The **layers button** (top right) turns on **Slope** (great for bluff lines), switches basemaps, and shows the **FEMA flood map**.
- **Open 3D WebGL Fly-Through** shows the terrain and every result in 3D.

2. WebGL Renderer

Hillshade and Slope are now separate layers - toggle each on/off from the layers button (top right), the same menu as the basemaps, so they can be combined.

Azimuth 315°

Altitude 45°

Z-Exaggeration 3x

☐ Multi-Directional Lighting

Terrain Smoothing Off (raw data)

No coarse-source pattern found (native-resolution LiDAR). Smoothing removes the flat triangle facets that sparse ground returns under forest leave in LiDAR DEMs, while keeping real breaks like bluff edges sharp.

Blends 4 lights spread around the compass (still anchored to the Azimuth slider) instead of just one, so a mound's flank facing away from the light isn't underlit or missed by eye - useful for visually sanity-checking scan results, since a real candidate should look like a coherent bump or depression from every side, not just the one the sun happens to be hitting. The classic single-light look (better for a strong sense of 3D form) is still there with this off.

Hillshade Opacity 85%

Slope Opacity 70%

Open 3D WebGL Fly-Through

Azimuth/Altitude/Z-Exaggeration only affect Hillshade. Slope shows true terrain angle (green=flat to dark red=60°+), deliberately unaffected by Z-Exaggeration - useful for spotting bluff lines and cliff faces, e.g. where a horizontal cave entrance might be, especially layered under a faint Hillshade.

3. Feature Detection

Phase 2: Long-Term Habitation (Base Camps)

Phase 2: Long-term habitation. Shelters that could carry families through a hard winter: large south/southeast-facing bluff overhangs (passive solar heat) with a flat, dry dirt bench at their base, near year-round water and above the 100-year floodplain.

Auto-Detect Best Settings for This Area

☒ Auto-detect after each download

Detected (0.7 × 0.7 km): LiDAR noise ±0.02 m, relief 54 m, median

Section 2: lighting, smoothing, opacity and the 3D view.

2.5 Scanning

- 1 In **3. Feature Detection**, choose the mode, e.g. *Native Mounds*. A short description of the mode appears under it.
- 2 Settings are auto-detected after each download (the box shows what was measured and *Settings applied*). Press **Auto-Detect Best Settings for This Area** to redo it.
- 3 Adjust **4. Detection Thresholds** if you want (see 3.1). For mounds, leave **Mound Shape** on *All Shapes*, or on *Learned from my confirmed mounds* once it appears.
- 4 Press **Scan Downloaded DEM**, or **Scan All (every mode)** to run all eight modes; **Stop Scan All** stops after the current mode.

On very large downloads (over 16 Mpx) a scan covers the part in view. Pan and scan again for the rest; there's no need to download again.

Native Mounds ▼

Earthen mounds from a multi-scale Local Relief Model.

Auto-Detect Best Settings for This Area

☒ Auto-detect after each download

Detected (0.7 × 0.7 km): LiDAR noise ±0.02 m, relief 54 m, median slope 3° (steepest 2%: 13°+), typical bluff 5.7 m, largest stream in view drains 67 ac (only drainage that falls inside the download is counted).

► **Settings applied (19)**
Mound shape presets are unchanged. Adjust any slider by hand if needed.

4. Detection Thresholds

Min Depth/Relief 1ft 6in

Min Circularity 0.15

Max Circularity 1.00

Min Diameter 15ft 0in

Max Diameter 850ft 0in

Mound Shape

All Shapes (recommended - runs Conical + Platform both) ▼

Min template match 0.60

Min height : width 0.030

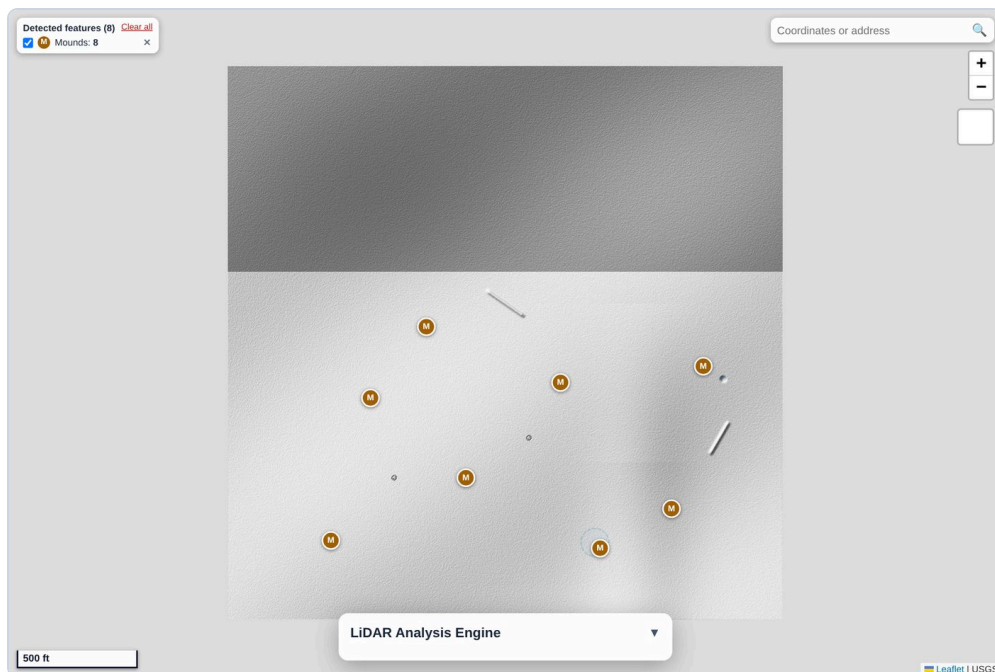
Error runs (persistence) 12

Min persistence 0.70

Method: iMound (Freeland et al. 2016). The terrain is detrended and flipped upside down, and the "pits" a fill has to raise are the mounds, measured from their own base contour, so only closed highs count, not hillside bulges or ridge noses. Each candidate is then re-tested with random 3DEP-grade LiDAR error added (stochastic depression analysis) and must persist in most runs. Finally it must correlate with an ideal conical dome or flat-topped platform (template matching, Davis et al. 2018).

Flat-ground slope 8°

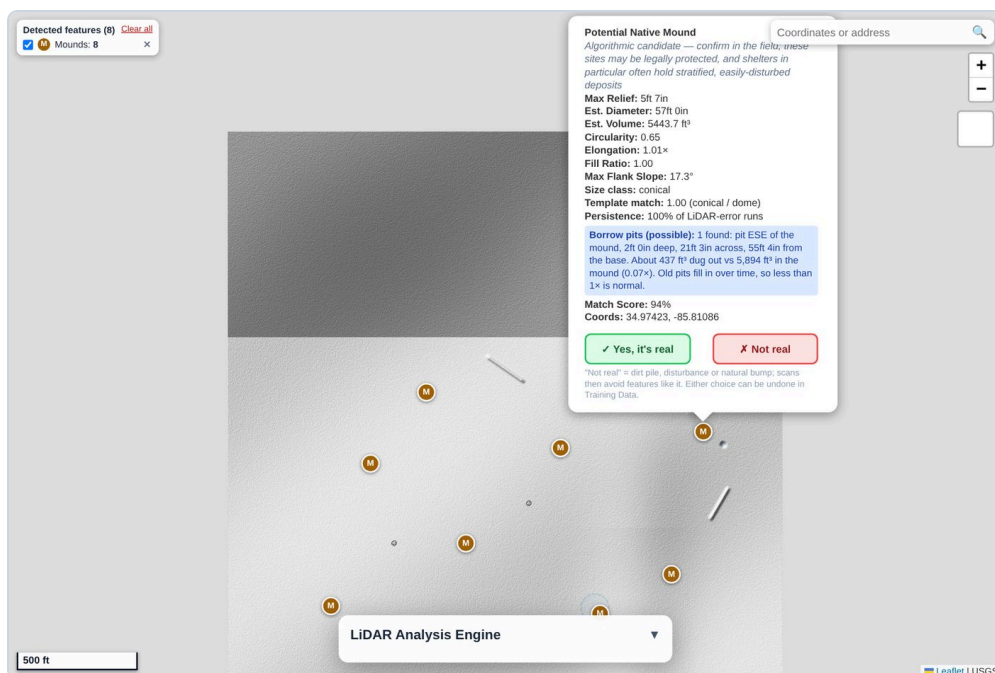
Sections 3 and 4: the mode, auto-detected settings and thresholds.



Mound scan results: brown "M" markers, counted in the legend at the top left. The panel folds away so the map is clear.

2.6 Reading and answering results

Tap a marker. The popup shows what was measured (height or depth, diameter, volume, roundness, stretch, flank slope, size class, template match, persistence), any warnings, borrow pits, similarity to things you've confirmed or rejected, and the **Match Score**.



A mound result with a borrow pit found beside it, and the two answer buttons.

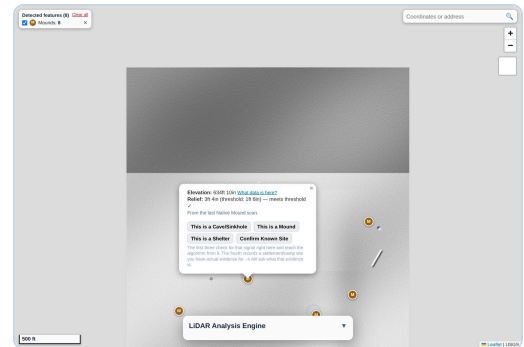
- **✓ Yes, it's real** — saves it as a confirmed example. A confirmed mound becomes a permanent waypoint (green ✓) that every later scan shows with the size recorded.
- **✗ Not real** — for a dirt pile, disturbance or natural bump. The result disappears; scans then skip that spot and skip or rank lower anything that looks like it (see **Avoid look-alikes**, 2.9).
- Not sure? Leave it unanswered. Wrong answers mislead the app more than missing ones, and any answer can be undone.

The legend's checkboxes show and hide each result type; ✗ clears that type; *Clear all* clears everything. Answers are kept even when results are cleared.

2.7 Checking a spot by hand

Tap anywhere on the terrain to see the elevation and the last scan's value at that point. The buttons check for a feature right there and save it if found:

- **This is a Mound** runs the full mound check at that spot and shows which tests pass or fail, with the mound outlined in brown. Use it when you know a mound is there but the scan missed it.
- **This is a Cave/Sinkhole** and **This is a Shelter** do the same for those features.
- **Confirm Known Site** records a camp or village site you have evidence for (it asks what the evidence is); it raises nearby and similar areas in the settlement scan.



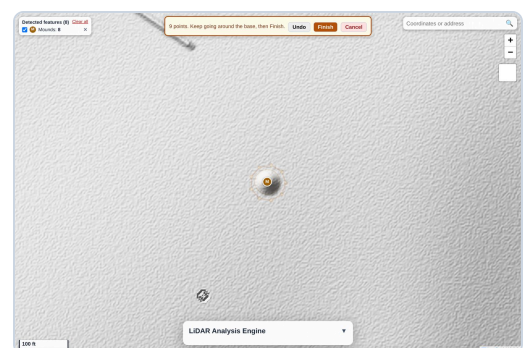
Tapping the map: elevation, relief, and the check buttons.

2.8 Outlining mounds and look-alikes

The most accurate way to record a mound you know, and the best way to teach the app what isn't a mound.

- 1 In **5. Training Data**, press **Outline a Mound or Look-alike**. Zoom in close.
- 2 Tap points around the **base** of the mound, where it meets the natural ground. **Undo** removes the last point.
- 3 Press **Finish**. The app measures it against the ground just outside your line and tells you whether your line looks tight or wide.
- 4 Save it as a **mound**, or as a **look-alike** (tree throw, boulder, spoil pile, natural knob, road or berm).

Show outlines, **Export outlines** (GeoJSON) and **Import outlines** (merges) are below the button.



Drawing an outline around a mound's base.

2.9 Teaching the scan your mounds

- **Learned settings.** After 5 confirmed mounds, **Mound Shape** gains *Learned from my confirmed mounds*, chosen automatically the first time. It sets the size and shape sliders to ranges that fit your mounds, with a safety margin that tightens as you add more. Confirm a variety (small and large, conical and platform).
- **Avoid look-alikes** (Training Data panel) sets how closely a candidate must resemble something you marked "Not real" to be skipped: Light 90%, **Normal 80%**, Strict 65%, or Off (rank lower only). Use Strict if similar false alarms keep appearing.
- Similarity to a confirmed real feature always wins a tie, so confirming real mounds of a size protects them from being dropped as look-alikes.

The Training Data panel shows how many examples of each kind you have, how many confirmations aren't backed up yet, the **possible issues** warning (2.11), and every training tool:

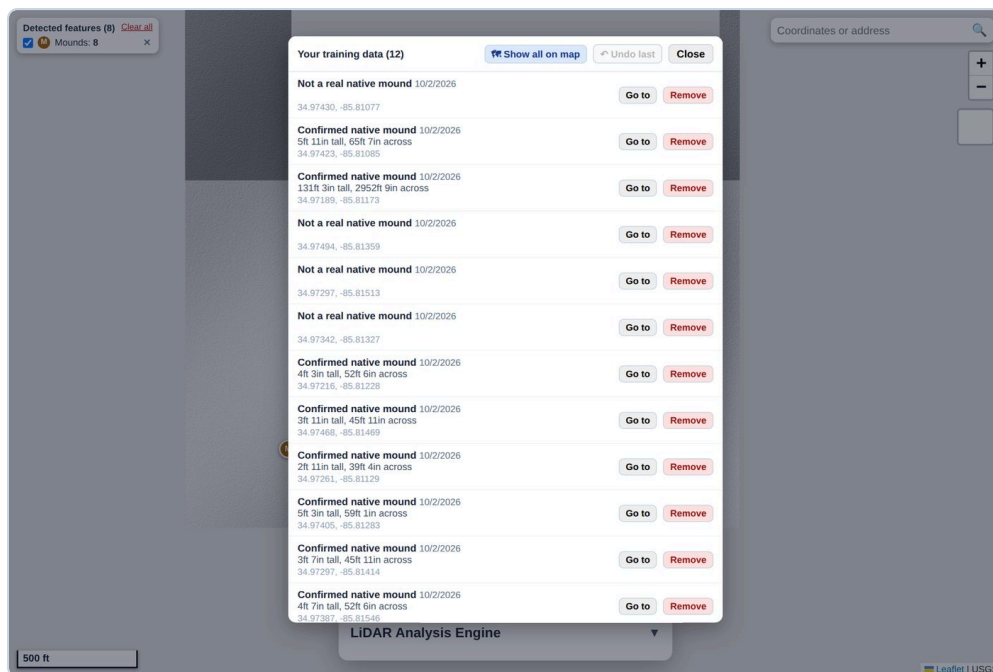
- **Undo last training change** — one step at a time, up to 30.
- **Review Training Data** and **Show Training Data on Map** (2.10).
- **Check training data for issues** (2.11).
- **Export** / **Import** / **Clear Training Data** (2.14).

The screenshot shows the 'Training Data' panel. At the top, it lists confirmed features: 0 base camp, 0 hunt camp, 0 ceremonial karst, 0 cave entrance, 0 cave/sinkhole, 8 mound, 0 shelter, 0 settlement example(s). Below this, a message states '1 confirmation not in a backup file yet. Back up now'. The 'Avoid look-alikes' dropdown is set to 'Normal (skip at 80% similar)'. There are buttons for 'Undo last training change' and 'Review Training Data'. A yellow warning box indicates '2 possible issues in your training data (2 probably wrong)' with a 'Review' link. Below this are buttons for 'Check training data for issues', 'Show Training Data on Map', 'Export Training Data', 'Import Training Data', and 'Clear Training Data'. A section titled 'Outline a Mound or Look-alike' shows 'Outlined: 0 mounds (0 the scanner can see), 0 look-alikes' and buttons for 'Show outlines', 'Export outlines', and 'Import outlines'. At the bottom, there is a detailed text explanation of how the app works: 'Tap points around the base of a known mound, where it meets the natural ground, then Finish. The app fits the natural ground in a band outside your line and measures height, size, shape, sides, ground slope, elevation and borrow pits. It also records what the scanner itself sees there, which is what later scans are compared with, and tells you if your line looks tight or wide. Outline look-alikes too (tree throws, boulders, spoil piles, natural knobs): later scans rank candidates like your mounds higher, rank ones like your look-alikes lower, and skip anything inside a look-alike outline. Export saves GeoJSON (opens in GIS software); import merges, so outlines from different trips and devices combine. Confirming a cave/sinkhole or mound — the checkbox on a marker's

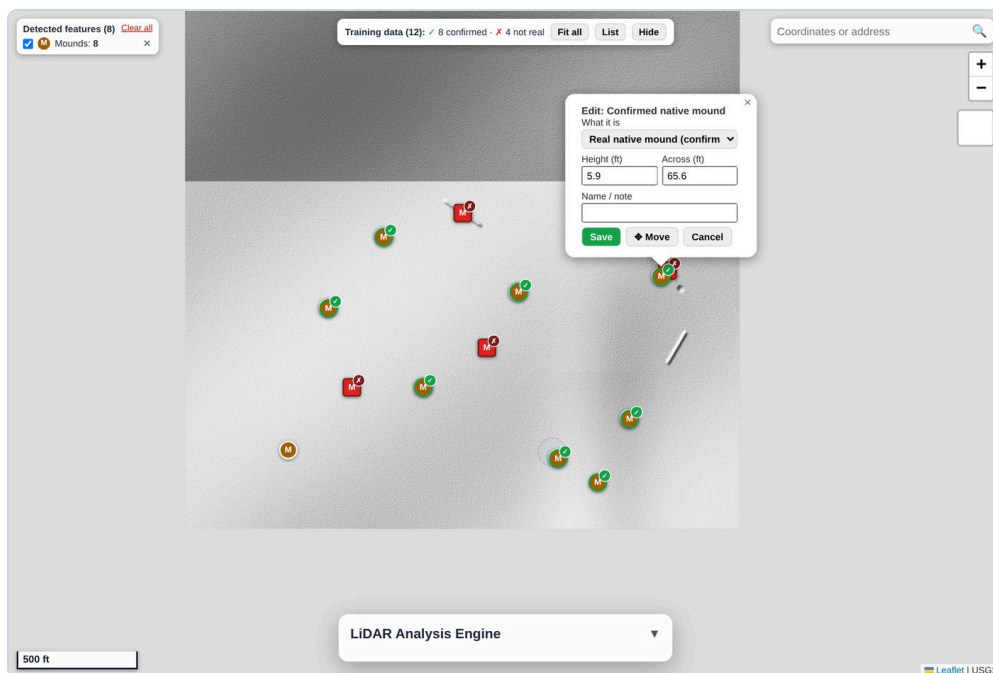
Section 5: Training Data.

2.10 Reviewing and editing training data

Review Training Data lists everything newest first, with **Go to** and **Remove**. **Show all on map** at the top (or **Show Training Data on Map** in the panel) puts it all on the map at once.



The training data list.



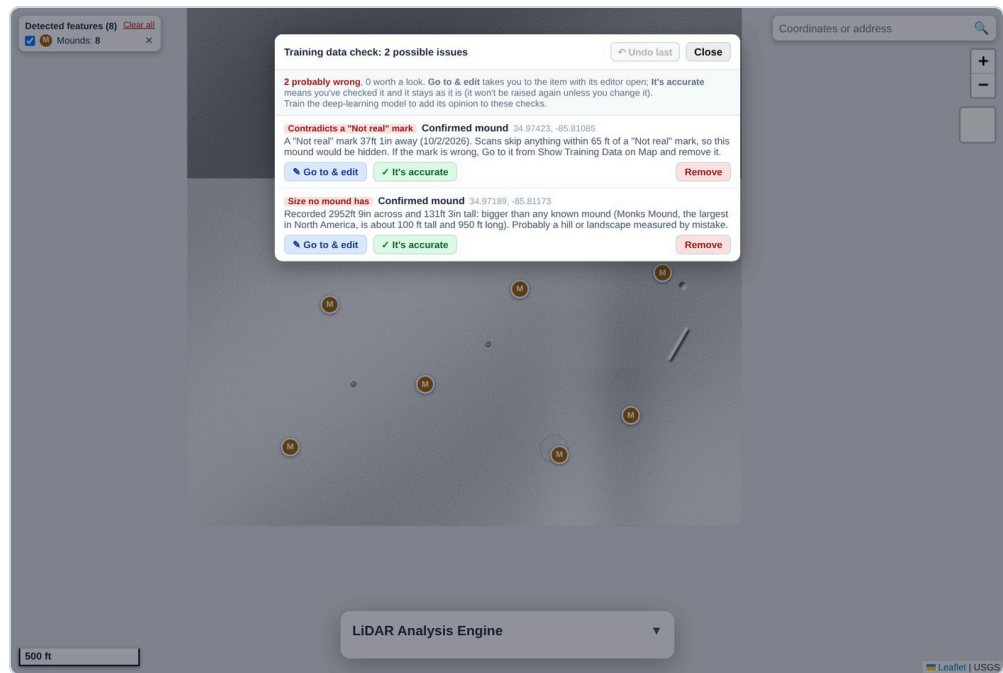
All training data on the map: confirmed (green ✓), not real (red X). Tapping one and pressing Edit opens its editor.

- 1 Tap an item for its details, then **Edit**, **Zoom in** or **Remove**.
- 2 In the editor, change **What it is** (real ↔ not real; mound ↔ look-alike for an outline), **Height** and **Across** in feet, or the **Name / note**, then **Save**.
- 3 **Move** lets you drag the marker to where the feature really is (an outline moves with it).
- 4 The bar at the top has **Fit all**, **List** and **Hide**. Every change is one Undo step.

2.11 Checking training data for issues

One wrong record — a hill saved as a mound, the same mound confirmed twice, a "Not real" tapped by mistake — quietly misleads both the scans and the network. When the app finds possible problems, the Training Data panel shows **⚠ N possible issues ... Review**. Press **Review** or

 **Check training data for issues**.

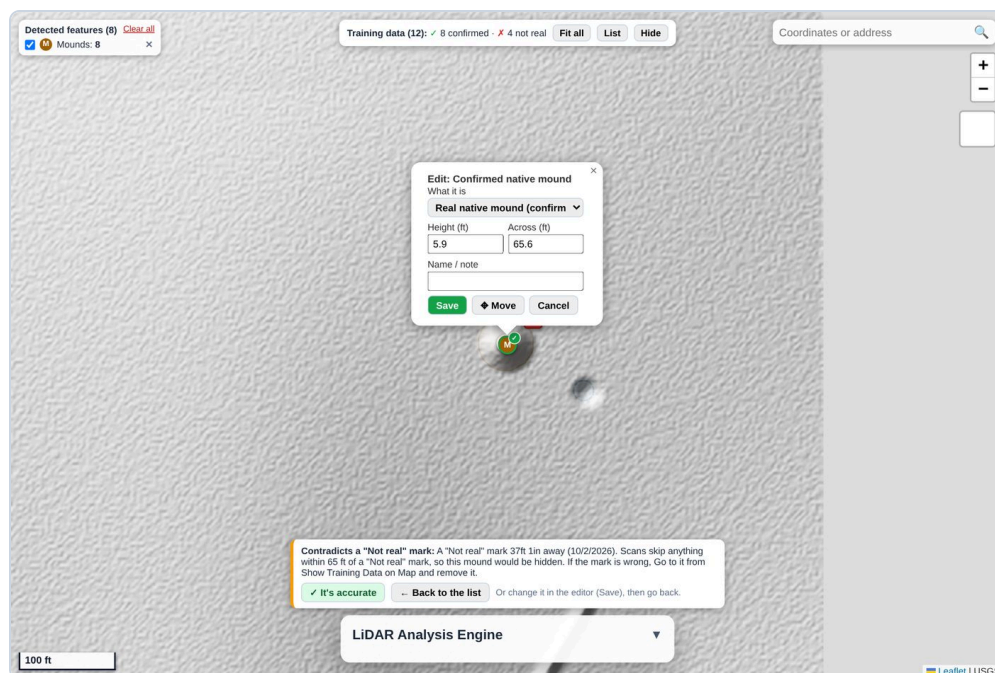


The check lists each possible issue, most serious first.

Issue	Meaning
Size no mound has	Over 1,150 ft (350 m) across or 105 ft (32 m) tall — usually a hill or landscape measured by mistake.
No size recorded	Saved without a height or width; outline it or enter the size.
Impossibly steep	Taller than half its width — mis-measured.
Very low / very small	Under 6 in (15 cm) tall or 10 ft (3 m) across.
Terrain doesn't match	The ground under the point barely rises; the point may be off the mound.
Confirmed twice	Two confirmations within 16 ft (5 m).
Contradicts a "Not real" mark	A confirmed mound within 65 ft (20 m) of a "Not real" mark — scans would hide it.
Network doubts / looks like a mound	With a trained network: a confirmed mound it scores under 30%, or a "Not real" mark it scores 90%+.

1  **Go to & edit** takes you to the item on the map with its editor already open, and a bar explaining the issue.

- 2 Fix it in the editor and **Save** — or press **✓ It's accurate** if the data is right. That issue won't be raised again for that item unless you later change its position, size or label.
- 3 **← Back to the list** returns to the check, which updates as you go. **Remove** deletes a record; **↶ Undo last** reverses any of these.



Go to & edit: the mound's editor is open and the bar explains the issue, with It's accurate and Back to the list.

2.12 Deep learning, start to finish

Section **6. Experimental: Deep Learning** trains a small neural network on your own training data. It works alongside the regular scan, not instead of it. Expect rough results until you have about 20 confirmed mounds.

Collect terrain samples

- 1 Load or download a DTM, then press **Collect samples from this DTM**. From then on, every DTM you load adds samples automatically.
- 2 The panel shows how many mounds and not-mounds have samples, and how many still need one. Press **Show them on the map**: they appear as orange (mound) and grey (not-mound) ? markers grouped into download-sized areas. Press **Download this area**; the bar moves to the next area until every item has a sample.

Train

- 1 Press **Train model** (needs internet the first time, to load TensorFlow.js). With a graphics card it takes a minute or two.
- 2 Read the result: the **held-out check** says how many mounds it found and how many not-mounds it wrongly flagged among examples it didn't train on. It also reports possible issues in your training data — fix them (2.11) and train again.

A small neural network learns what mounds look like from **your training data**: confirmed mounds and mound outlines teach it what a mound is; "Not real" marks, look-alike outlines and random ground teach it what isn't. It looks at the bare-earth terrain directly (40 m squares, ground tilt removed) instead of following the scan's rules, so it may catch mounds the rules miss, or flag things they wouldn't. It needs a terrain sample of each item, which is taken whenever a DTM covering it is loaded. More mounds = better: expect rough results under ~20 confirmed mounds. Results are separate from the regular scan (purple "D" markers).

Samples: 7 mounds, 4 not-mounds, 150 background (1 area)
1 mound wider than 115 ft is left to the regular scan (too big for the network's 131 ft window).

Model: trained 10/2/2026, 12:06:11 PM on 7 mounds and 154 not-mounds; held-out check: found 1 of 1 mounds, wrongly flagged 0 of 31 not-mounds

Confidence **80%+** ▼

Collect samples from this DTM

Train model

Check training data

Scan with deep learning (area in view)

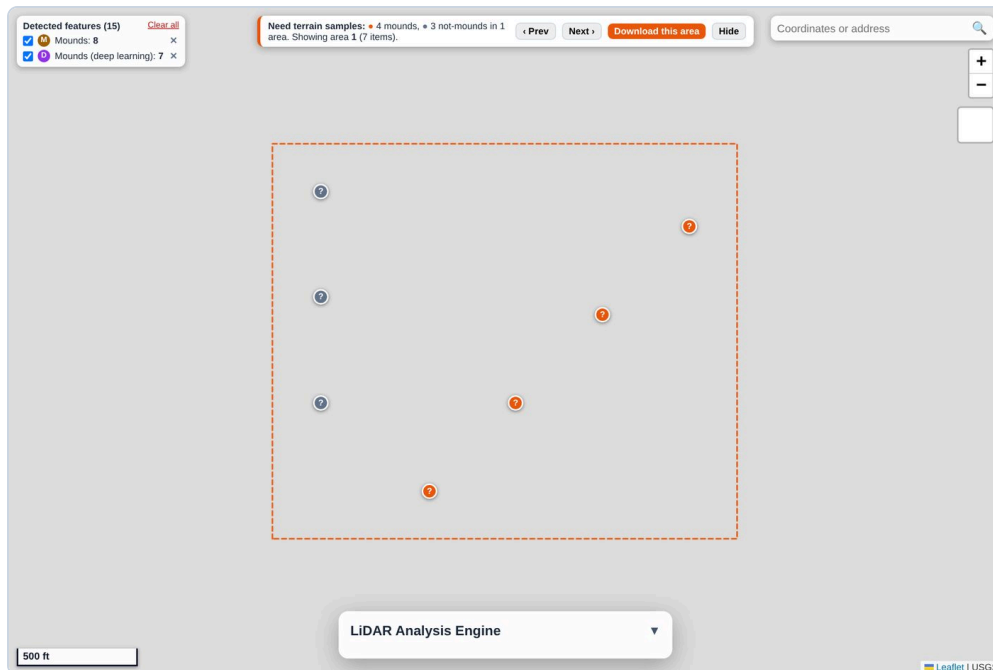
Trained in 51 s on 7 mounds and 154 not-mound samples (1 mound too big for its window left to the regular scan). 4 possible issues in your training data: press Check training data. Held-out check (data it didn't train on): found 1 of 1 mounds; wrongly flagged 0 of 31 not-mounds. Now press Scan with deep learning.

Save deep-learning model **Load deep-learning model**

Saves one file with the trained model, its terrain samples and your training data. Load it to restore everything here or on another device; loading adds to what is already here.

Delete model and samples

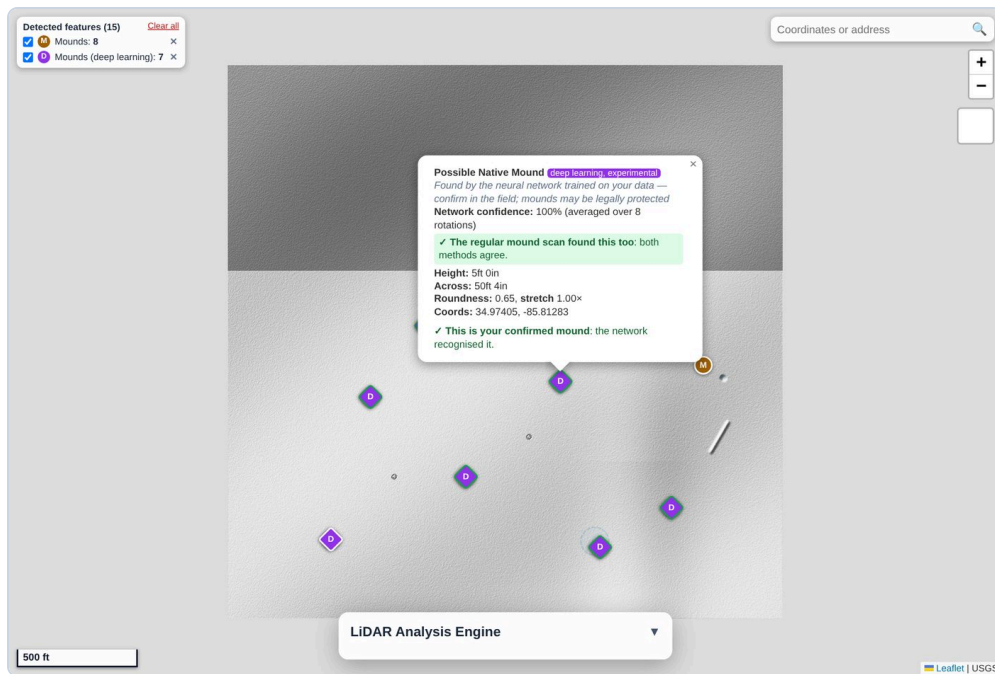
Section 6 after training.



Items still missing a terrain sample, grouped into an area with Download this area.

Scan and answer

- 1 Run the regular **Native Mounds** scan first (for the cross-check), then zoom to up to about 1 square mile and press **Scan with deep learning (area in view)**. Choose **Confidence** (80% default; lower shows more with more false alarms).
- 2 Results are purple **D** diamonds on their own legend line. Popups show the network's confidence (averaged over 8 rotations), the measured size, and **"The regular mound scan found this too"** when both methods agree — those, and your confirmed mounds, are listed first. Candidates smaller than your smallest mound setting are flagged and listed last.
- 3 Answer **✓ Yes, it's real** or **✗ Not real**. Answers go into the same training data the regular scan uses. Train again after every 5–10 answers.



Deep-learning results (D) with a candidate that both methods agree on and that is already a confirmed mound.

Mounds wider than 115 ft (35 m) don't fit the network's 131 ft window and are left out of its training; the regular scan handles big platform mounds. Without a graphics card training is very slow; scanning falls back to a faster engine.

2.13 Exporting for the field

- **GPX:** type a file name and press **Export GPX**. Every detected feature of every type becomes a waypoint with a type-specific symbol, a lettered name (e.g. *E03 Cave entrance setting*), elevation, score and size. On desktop Chrome/Edge a Save dialog lets you choose the folder.
- **Export Candidates (GeoJSON)** and **Export Shapefile (.zip)** for GIS software; confirmed mounds and outlines export with their shapes.
- **Export Analysis Raster (GeoTIFF)** saves the last scan's measurement grid.

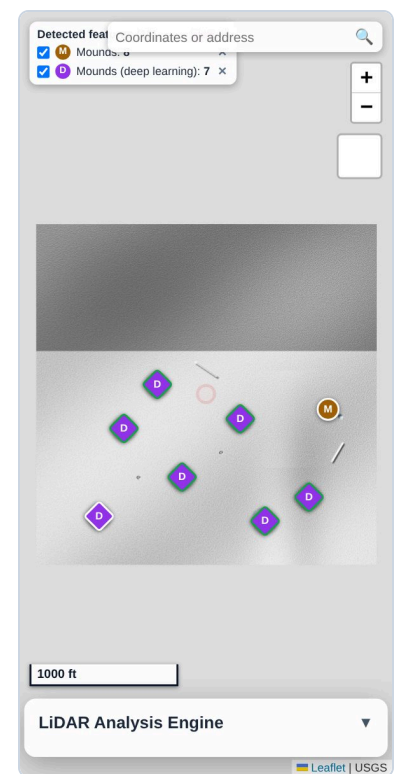
2.14 Backing everything up

Everything is saved automatically in your browser, but clearing the browser's site data erases it. Keep files:

- **Export Training Data** after each session (or **Back up now** when the panel says confirmations aren't backed up). Where the browser supports it, *Keep a backup file updated automatically* rewrites a file after every confirmation.
-  **Save deep-learning model** saves one file with the network, all its terrain samples and a copy of your training data.  **Load deep-learning model** restores it here or on another device.
- Importing or loading **merges**: nothing newer is lost and nothing is duplicated.

2.15 Using it on a phone

- The panel folds to a bar at the bottom; tap it to open. It folds itself away when you search, draw or review on the map.
- Search is the  button at the top right.
- Use a **Memory Budget** of 16–32 Mpx and download smaller areas.
- Local USGS tiles work: only the part in view is read.
- Popups' Yes / Not real buttons are large and far apart to avoid mis-taps.



Phone layout.

2.16 Batch pipeline (desktop)

For large areas, `pipeline/karst_pipeline.py` runs the same phases with GDAL/Rasterio:

```
pip install -r pipeline/requirements.txt
python pipeline/karst_pipeline.py my_dtm.tif --out results/ --caves known_caves.geojson
```

Open the resulting `dtm_wgs84.tif` with **Load Local DTM** and `all_candidates.geojson` with **Load Pipeline Results** to review them in the app and in 3D.

Reference

3.1 Settings reference

Shared thresholds (section 4)

Min Depth/Relief	The smallest height (or depth) a feature must have.
Min / Max Circularity	How round a feature must be (1 = a circle; a square is about 0.79).
Min / Max Diameter	The size range of features to keep.
Exclude Waterways / Waterway Threshold	Drops candidates on stream channels draining more than the threshold area.
Max results per mode	How many of the best-scoring candidates are shown.

Mound settings

Mound Shape	All Shapes (Conical + Platform, recommended), Conical, Platform, Broad/Any (quick), Custom (sliders), Learned (from 5+ confirmed mounds).
Min template match	How closely a candidate must resemble an ideal dome or platform (0–1).
Min height : width	Rejects broad, barely-raised swells.
Error runs / Min persistence	How many times a candidate is re-tested with LiDAR error added, and in what share of runs it must survive.
Flat-ground slope	Above this ground slope (8° default) a candidate needs a clean mound shape to count; nothing on ground over 20° is kept.
Borrow pits	Look for them (rank mounds with pits higher), show only mounds with pits, or don't look.
Trend Radius, Flank Slope Warning, Elongation Warning	Custom-mode detrending size; warnings (not exclusions) for unusually steep or stretched candidates.

Other modes

Base camps (Phase 2)	TPI radius and extreme, minimum bluff slope; aspect window (135°–225°); bench slope, distance, length and midden flag; water buffer, year-round stream size, floodplain (HAND) cutoff.
Hunt camps (Phase 3)	Outcrop TPI radius and slope; choke-point scale, distance and canyon width; viewshed radius, observer height, minimum visible valley floor, and whether a viewshed is required.

Ceremonial karst (Phase 4)	Sinking-stream drainage; only sinks fed by sinking streams; cave survey file, match distance and dark-zone cave length.
Cave entrances	Cliff and apron slopes, distance to cliff, cove search radius, stream-head distance, sinks-upslope distance, preferred aspect, and whether a cove or spring head is required.
Caves / sinkholes	Minimum depth-to-width and wall slope; creek pools shallower than the cutoff are dropped.
Rockshelters	Search radius, minimum cliff slope, maximum floor slope.
Settlement likelihood	Weights for water, flood safety, flat ground and known sites, plus decay distances, flood-safety height range, slope sensitivity, neighbourhood scale and the top % highlighted.

3.2 Troubleshooting

Problem	What to do
"Over the memory budget"	Zoom in, choose a coarser resolution, or a larger budget (desktop only above 64 Mpx).
Scan is blocked (coarser than 1 m)	Use the original USGS 1 m tiles with Load Local DTM, or untick <i>Require ≤ 1 m/px</i> (results on coarse data are unreliable for small features).
A mound you know wasn't found	Tap it and use <i>This is a Mound</i> to see which check fails, or outline it. Consider <i>Learned</i> settings or lowering Min template match.
Too many false alarms	Mark them Not real; set Avoid look-alikes to Strict; confirm real mounds of the same size so they aren't dropped.
Training data disappeared	Open the app from the same address as before (including the port). Import your last backup; importing merges.
Deep learning: "TensorFlow.js could not be loaded"	It loads from the internet on first use. Connect once; browsers usually keep a cached copy afterwards.
Deep learning is very slow	The device has no usable graphics card. Training will take long; scanning uses a faster fallback. Train on a computer, Save the model, and Load it on the phone.
Scale bar or buttons hidden on a phone	Update to the latest index.html; the map now fits the visible screen above the browser's toolbar.

3.3 Responsible use

Every result is an **algorithmic lead, not a confirmed site**, and needs checking in the field. Mounds, rockshelters, caves, burials and mud-glyph sites are protected by federal law (ARPA, NAGPRA) and state cave and burial laws. Don't disturb sites, get landowner permission, and keep coordinates and cave-survey data confidential — including the GPX, GeoJSON and backup files you export.